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Investigation Into Daily Impact of Transportation Services

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Project Summary

This study is interested in the effects of transportation on individual quality of life. It includes a review of the literature concerning transportation in recent times, the research proposal, and details of this study. This study will investigate how students at the University of Washington interact with local transportation services and attempt to identify direct impacts on their out-of-class lives. This proposal discusses some of the challenges in attempting to find this information, including the variety of extraneous variables that influence a student's experiences.

Intellectual Benefits

This study attempts to find provide a Geographer's perspective to the subject of transportation. This is a subject that has been extensively studied by economists, urban developers, policy makers, and those investigating societal equity. Comparatively little has been done to examine the unique and deeply personal implications of transportation access on individual livelihoods. This study seeks to describe those personal interactions, and develop our understanding of transportation affects the individual.

Practical Policy Benefits

The practical benefits of public transportation services are increasingly well understood. It is a useful means-to-end to help people travel to places they otherwise couldn't. The results of this study would enhance our understanding of the depth to which access to transportation impacts the day-to-day choices of individuals. It would provide indication of whether transportation options merely serve mandatory obligations, or whether they can liberate an individual's personal life. This would provide a new layer of context for transportation planners and policy makers. Based on this information, they could examine the appropriate level of investment to allocate for developing and maintaining transportation services. Do they need only maintain mainline routes that touch important centers? Or is there further value in expanding smaller and more niche routes as well.

Literature Review

Opening statements:

Discussions about transportation, whether public or private, are a natural topic for geographers. The explosion of economic growth and personal mobility experienced within the last century has been the topic of many geographic studies (Keeling, 2008). Many of these studies investigate the accessibility of transportation amongst groups of individuals, with economic status consistently a common subject (Guzman, Ovideo, and Rivero 2017; Mora, Truffello, and Oyarzún 2021). The term 'accessibility' comes frequently in any discussion of transportation and geography. Geurs and van Wee (2004) popularized the notion of accessibility as the combined definition of four elements: land use, transportation, temporal, and the individual. Together these four elements represent the spatial distribution of a transportation system. The individual is a set of qualities and descriptions about the person who uses transportation. This can include their level of income, level of education, location of home, and other such considerations (Geurs and van Wee 2004). Land use constitutes the origin, path, and destination of the traveler, while the temporal deals with time they spend traveling. Transportation covers the specific mode of transport used by an individual.

Economic Geography of Transportation:

From this basic definition of accessibility comes a host of inquiry into the equality of transportation. As mentioned, the accessibility of transportation services to individuals of various economic backgrounds is a prevalent conversation (Deboosere and El-Geneidy 2018; Guzman, Oviedo, and Rivero 2017; Mora, Trufello, and Oyarzún 2021). The accuracy of the metrics and data used by city planners and developers to organize efficient and available transportation were the subject of Deboosere and El-Geneidy (2018). In researching the parameters used by planners in Canada, the pair investigated how accessibility to employment was measured. They found and subsequently argued that simply considering all jobs accessible by public transportation was not indicative of the true accessibility experienced by lower versus upper income workers. Their reasoning was that not all jobs are worked by all workers, and that a true indicator of lower income accessibility should consider only the jobs normally occupied by those individuals, and similar for higher income individuals. On applying these revisions to their models and completing their study, they found that the true accessibility for lower income workers was significantly lower than the models used by city planners (Deboosere and El-Geneidy 2018). A study of cycling infrastructure in Santiago, Chile also revealed a bias in development towards wealthier sections of the city (Guzman, Oviedo, and Rivero 2017). Here, the researchers found that lower income residents were often forced onto crowded public transportation services and weren't afforded an equal level of cycling access as the wealthier sections of the city. They noted that similar scenarios were observed in North America as well.

The Human Factors:

While economic factors are a common focal point for transportation geography, other factors have also seen research. A study of walking accessibility in China (Cheng *et al* 2019) investigated the differences in accessibility for the elderly versus the younger generation. The study pointed out the increasingly aging portion of the world's population, and especially of China's population. As they noted, in China the elderly continue to participate in pastimes and hobbies outside the home like chess rooms, parks, and other public entertainment. Their access to these resources therefore remains an important component of daily life.

Many studies handle the accessibility of transportation as a necessary function for taking individuals to work, or to school, or to the stores. Many also delve into the rich connections between transportation and commercial activity. Cheng *et al* (2019) highlights a need for transportation that stretches beyond the economic and the purely functional. The ability to exit the home and travel can be relaxing, social, and even entertaining.

Impact on Human Life:

While economic and practical factors are important subjects for geographers to investigate and research to understand the quality of life of individuals, the other opportunities provided by transportation may be equally valuable. An area I think could be explored is the effect of quality transportation services on daily human experiences. Put another way, a

study of whether an individual's lifestyle choices are tied to the accessibility of transportation could be a fascinating look into the quality of modern human life. When an individual is comfortable with their ability to move about and travel, what kinds of opportunities do they explore? Conversely, does the lack of quality transportation inhibit a person's ability to pursue their interests and enjoy their surroundings? How would socialization amongst youths improve with efficient ways to meet at a favorite gathering place? What comfort does the ease of transport add to a maturing adult's interest in stepping outside the house? Investigating these relationships could assist in improving individual quality of life and strengthening societal connection across the country. Answering these questions regarding the general public would be too broad for a single study. For my purposes I would like to start with a case study of individuals who share similar traits, concentrated on one geographic place. For reasons I will elaborate on, I plan to look at students at the University of Washington.

Research Question

The question I wish to investigate is: **How does accessibility to transportation affect the out-of-class lives of students at the University of Washington?**

Methodology

Overall Research Design

This research initiative will be a Descriptive investigation. The goal is to develop an ideographic understanding of the interactions of UW students with local transportation options. I will use qualitative interviewing to gather perspective from students on the factors that impact their use of transportation. Based on the responses gathered, patterns and trends can be documented for future investigations.

Population

This study will focus on undergraduate students from the University of Washington. While the UW has a Bothell and Tacoma location, we will specifically be looking at students from the Seattle campus. As the largest of the three campuses, UW provides both the largest sample opportunity as well as the widest variation between on-campus residents and commuters. The Seattle campus also enjoys proximity to the many different routes for bus, rail, streetcar, and pedestrian travel options for Seattle itself. UW Seattle students are also automatically enrolled in U-Pass every quarter as part of tuition. This provides consistent availability of transportation across all students.

University students tend to be highly mobile. They regularly travel within campus to classes, study groups, or clubs, and often travel beyond campus grounds for personal recreation and events. With such widespread mobility, a likely significant portion of them

will use transportation to move about. By studying the movements of university students, we can find useful information about the effect of transportation on their lifestyle.

Unit of Analysis

The unit of analysis for this study are the individual students at University of Washington Seattle, and the relationship between their usage of transportation services and the quality of life they enjoy outside of the classroom.

Unit of Observation

My unit of observation will be the testimonies of interviewed University of Washington students. After concluding the interviews, I will use the information they provide to draw connections and patterns that highlight certain factors affecting their experiences. Notable items include frequency of transportation usage, the distances travelled, and overall satisfaction with the journey. These will indicate the students' level of satisfaction with transportation.

Sampling Strategy

Due to the inductive and qualitative nature of the study, I will use a Non-probability Sample. The raw number of participants is not as important as taking information from a variety of individuals. Obtaining even a few commuters in the data collection could introduce the perspective on non-residents and their experiences with transportation.

More important to the study is taking interviews with a variety of individuals. Accessing on-campus residents would be simpler than accessing off-campus commuters. For on-campus residents, the dormitory / apartment system and the presence of Residential Advisors provide a secure and soft entry point. RAs could be asked to share my contact information with their students, who would have the opportunity to volunteer as a potential interviewee. The centralized nature of campus residency and familiarity between RA's and residents would work for me here.

To include off-campus commuters, the most practical method would be bulletins placed at strategic locations. These will include near UW parking lots and bus terminals. There is the possibility that UW Transportation Services would agree to send an email to registered commuters, alerting them to this study and the potential to volunteer. This would mostly target the students who drive to campus, and be less effective with those who walk, car share, or use other public transportation. In any case, the response rates for this would likely be lower than the response rates of residents. Since non-UW individuals would also have access to these bulletins, a vetting system is required to ensure the sample remains unique to UW students.

A solution that maintains the unity of the sample group would be an intermediary internet survey. Requested information would include their name, UW email address, class standing, current or intended major, and status as a commuter or on-campus resident. This survey would be linked in an email from the RAs and/or included in a QR code on bulletins. Vetted responses would be randomly chosen until the quotas are met (see Sampling Type section). The surveys could then be destroyed.

Sampling Type

The specific sampling technique will be a quota sample. I want to maintain an equal number of responses from each class year, or equivalently class standing. My prediction is that class standing will be relevant to the respondent's experiences, so I want to account for this with an equivalent spread of answers.

Another consideration is the balance of on-campus versus off-campus students, or residents versus commuters. For the reasons stated previously, I will likely find more on-campus volunteers than off-campus volunteers. However, I predict that the difference between commuting versus living on campus will outweigh the importance of class standing. Based on this, I don't believe it's necessary to enforce an equal number of commuters and residents in the study. This is especially true for freshman students, who (to the latest of my knowledge) are required to live on campus by default. To ensure the commuters are accounted for, at least 25% of the final interviewees will be commuters. To maintain the breadth of possible responses and the commuters' experiences don't overshadow the findings from the on-campus residents, no more than 50% of the final interviewees will be commuters.

Estimated Sample Size

The sample size must remain practical while still offering a rich base for information. I would aim for a 32-student sample, with eight students each of freshman, sophomores, juniors, and seniors. With a predicted pace of conducting two interviews a day, this would be 16 days' worth of interviewing. Of the 32 interviewees, at least eight must be commuters, but no more than sixteen. For each class year, no more than four of the eight interviewees should be commuters.

Mode of Observation

The mode of observation I have chosen is an in-person qualitative interview. There are some disadvantages with this choice. The time investment needed to conduct an in-person interview reduces the number of samples which can be taken, compared to a method like an internet survey. Scheduling between myself and the interviewees must be considered as well, since they should be available to speak for a suitable period without disrupting their studies or other plans. Logistics also includes notetaking and recording equipment, finding a suitable interview environment, and any add-ons to make the process more agreeable. I intend to provide hot chocolate for interviewees as a sign of gratitude and to make them comfortable. This requires budgeting and arrangements.

Yet for the purpose of examining quality of life, an intensive in-person qualitative interview does have advantages. The key upside is the depth of information obtained from each interview. An in-person meeting with well-curated questions should find high quality data

that fleshes out a full picture of a UW student's experiences with transportation. I can coax deeper explanations and motivations at times where a written response survey may return more hollow results.

List of Variables

One of the challenges in this study is distinguishing transportation-related variables from other unrelated factors that affect a respondent's experiences. Capturing these extraneous variables during the interview will highlight trends. We will then have a clearer understanding of the causation relationships.

One important variable to this study is the interviewee's residency status. This determines whether interacting with transportation is a requirement for that individual to be on campus at UW. There are three possible states for the student; they are on-campus residents; they are commuters; they are both commuters and on-campus residents. An example of the last state is a student who has a room in a campus residential hall, but some days will return home after class. A similar situation is for a student who returns home on weekends. For the qualitative nature of this study, any entry of "on-campus resident", "commuter" or "both" is valid and usable. For this reason, my interview script asks for their status as a resident and a commuter in two separate questions (see Appendix, Script Question #4 and #5). Any of these answers will yield useful information for learning about the students' interactions. The answers to this question will not be affected by the process of interviewing and will accurately reflect a real-world truth. For understanding their perspective and experience with transportation, their residency status is an independent

variable. The status of this variable is likely to affect their level of interaction with transportation. It is not, however, the only factor likely to influence their experiences with transportation. Therefore, other potential influences will also be recorded.

The class standing, or class year, of the student should be a control variable. To clarify, while I use the two terms interchangeably, the intention of this variable is the current class standing of the student based on their credits completed. For undergraduate students at UW this will have an answer of freshman, sophomore, junior, or senior. It is possible that differences in class standing may result in differences in transportation experience.

However, the sampling structure of this study is intended to keep class standing as a consistent baseline. An equal number of students from each class year will be included in the study. Relative to students of the same class standing, this variable should be constant. This variable will serve as one indicator for the workload (or perceived workload) of the student (see Appendix, Script Question #8). This variable is a standardized value that should accurately reflect the study and living conditions of the student.

Similarly, another variable in this study is perceived workload. This is referenced in Appendix, Script Question #8. Perceived workload refers to the respondent's opinion of the time commitment, stress, value, and/or difficulty of their course material. In this study it is an independent variable in this study. Though inherently subjective and open-ended, it nevertheless may give insight into the interviewee's perspective. If they believe their workload is heavy, versus if they believe their workload is lighter, how does this affect their willingness to travel off campus for non-academic reasons? To maintain the validity of their response, Question #6 precedes the other questions concerning employment, RSO

membership, and travelling experience (see Appendix, Questions #9 onwards). The goal is to avoid any self-conscious feelings from the interviewee prior to answering this subjective question. Depending on their answer to those other questions, I unwittingly might come across as belittling them for not showing signs of being busy. In this way I hope to avoid the Hawthorne effect, where the interviewee is sensitized by my previous questions.

Several of questions in my interview script address potential independent variables in quick succession. These include intended major, membership in an RSO, and employment status (see Appendix, Script Questions #2, #9, #10, #11). There are finite answers to these questions. Intended majors are standardized by the University of Washington. Membership in an RSO and employment status are either answered “Yes” or “No”. The location of employment will be “Not employed”, “Employed on campus”, or “Employed off campus”. In the event a student has a job on-campus and another job off-campus, the answer will default to off-campus employment. These variables are potential influences on a student’s willingness or ability to travel off campus, but unrelated to the issue of transportation itself. All of them should produce standardized, non-stimulated answers that accurately represent the students’ current activities.

The remainder of the questions in the interview script are open-ended, and delve deeper into the interactions between students and transportation. They allow the interviewee to give as detailed a description of their experiences as they wish.

Timeline

1. Complete intermediary survey and finalize the script for interviewing. Locate possible interview locations, will be finalized later
 - a. Should take 2 days maximum for survey
 - b. Script may take the same two days
 - c. Location searching may take two to three days to narrow down
2. With permission, post bulletins in frequented areas, with my email and QR code to survey
 - i. Alongside this, request permission to ask RAs to share my information and the same survey with their residents
 - ii. Request UW Transportation Services if they are willing to likewise share my information with registered commuters
- b. After posting these, wait at least 2 weeks to allow volunteers to submit
3. Once volunteer emails have come in, vet the individuals to ensure they are UW students. Randomly select participants to fill the quota requirements (eight per class year, eight to sixteen total commuters)
 - a. Keep two to three days to carefully review each candidate
4. Send out emails to finalists requesting an interview time and chose interview location. Give three-week window for flexibility or rescheduling. Also arrange for the hot chocolate.
 - i. If some finalists no longer able to commit, randomly select from unchosen candidates, still following quotas

- b. Allow an additional week for finalists to respond and select a time
5. Once dates settled, will be interviewing for next three week window. Plan for two interviews a day to allow full debriefing with interviewee and self-reviewing.
 - a. Plan for three weeks
6. Once interviews complete, examine data and begin to build report.

Total time estimate: ~48 days to complete interviews

Budget

BUDGET NAME	Overall Time Frame Month or Quarter 1	Month or Quarter 2	Month or Quarter 3	Month or Quarter 4	TOTALS
01-Salaries					
Include yours if you would be paid. Also include any research assistants you would need.	\$7,000	\$4,320	\$4,320		
~48 days, 8hr per day, \$18 hr		30 days, 8hr day, \$130 days, 8hr day, \$18			
02-Personal Service Contracts					
Do you need to pay anyone for their services (e.g. Translators, Transcribers, Consultants)	32*\$20				
Paying volunteers(?)					
03- Other Contractual Services					
Do you need to pay an outside company (e.g. for advertisements)	\$3 * 100 + \$40				
Laminating plus printing bulletins					
04 - Travel					
local travel, assume 0.25 cents per mile. Include airfare and accomodation.		\$120			
Traveling to campus ~20 days, in gas					
05 - Supplies and Materials					
e.g. Software, Hardware, Instruments, Postage, Office Supplies	\$50 + \$6 + \$10				
notepad, pen/pencil, hot choc packs					
06- Equipment					
Hardware, Instruments. (e.g. GPS)	\$70				
Voice recorder					
07 - Benefits					
Calcuate 14.2% of your salary. Otherwise consult A&S instructions	\$994	613.44	613.44		
08 - Student Aid and Other Grants and Services					
e.g. your tuition					
Total Direct Costs:					
Amount Subject to F&A Costs: Typically this would be amounts in 01 thru 06	9110	4933.44	4933.44		
Indirect Costs (UW Overhead) : multiply Total Direct Costs by 54.5% to get total budget amou	10342.4				
Total Total Amount of Budget:	29319.28				

Appendix

Script Questions:

The script will be a printout sheet of the following question, which I will keep with me throughout the interview.

- 1) What is your first and last name?
- 2) What is your currently intended major?
- 3) What is your current class standing?
- 4) Do you live on campus?
- 5) Do you commute to campus?
- 6) If you commute to campus, do you use private transportation?
- 7) If you commute to campus, do you use public transportation?
- 8) Which of these descriptions is the closest match for your current school workload
(from greatest to least): Highly stressful, Difficult, Manageable, Light, Negligible
- 9) Are you part of an RSO (Registered Student Organization)?
- 10) Are you currently employed?
- 11) If employed, are you based on campus, off campus, or under some other conditions?
- 12) How would you describe your activities on-campus when not in class?
- 13) If you have traveled off campus during the school year, how would you describe the frequency of your travels?

14) Tell me about your experiences traveling off campus during the regular school year.

Off campus means non-UW UW facilities and sites.

15) If you have traveled off campus during the school year, what are some places you've visited? (If they don't want to share specific places, then general locations).

16) If you have traveled off campus during the school year, how did you travel to locations?

17) Were you satisfied with your journey experiences?

18) Are there things about transportation in Seattle that you wish were changed?

19) If you haven't traveled off campus during the school year, what are the reasons?

Other Tools

Interviews will be recorded on paper notebooks by handbook.

With permission from the interviewee, they will also be recorded with a voice recorder, such as those available here:

<https://www.amazon.com/s?k=voice+recorder&crid=M08GEKWWA8SU&srefix=voice+recorder%2Caps%2C247>

Bibliography

1. Cascetta, E. et al. (2020) 'Economic growth, transport accessibility and regional equity impacts of high-speed railways in Italy: ten years ex post evaluation and future perspectives', *Transportation Research Part A: Policy & Practice*, 139, pp. 412–428. doi:10.1016/j.tra.2020.07.008.
2. Cheng, L. et al. (2019) 'Investigating walking accessibility to recreational amenities for elderly people in Nanjing, China', *Transportation Research Part D: Transport & Environment*, 76, pp. 85–99. doi:10.1016/j.trd.2019.09.019.
3. Deboosere, R. and El-Geneidy, A. (2018) 'Evaluating equity and accessibility to jobs by public transport across Canada', *Journal of Transport Geography*, 73, pp. 54–63. doi:10.1016/j.jtrangeo.2018.10.006.
4. Geurs, K.T. and van Wee, B. (2004) 'Accessibility evaluation of land-use and transport strategies: review and research directions', *Journal of Transport Geography*, 12(2), p. 127. doi:10.1016/j.jtrangeo.2003.10.005.
5. Guzman, L.A., Arellana, J. and Alvarez, V. (2020) 'Confronting congestion in urban areas: Developing Sustainable Mobility Plans for public and private organizations in Bogotá', *Transportation Research Part A: Policy & Practice*, 134, pp. 321–335. doi:10.1016/j.tra.2020.02.019.
6. Guzman, L.A., Arellana, J. and Castro, W.F. (2022) 'Desirable streets for pedestrians: Using a street-level index to assess walkability', *Transportation*

Research Part D: Transport & Environment, 111, p. N.PAG.

doi:10.1016/j.trd.2022.103462.

7. Guzman, L.A. and Oviedo, D. (2018) 'Accessibility, affordability and equity: Assessing "pro-poor" public transport subsidies in Bogotá', *Transport Policy*, 68, pp. 37–51. doi:10.1016/j.tranpol.2018.04.012.
8. Guzman, L.A., Oviedo, D. and Rivera, C. (2017) 'Assessing equity in transport accessibility to work and study: The Bogotá region', *Journal of Transport Geography*, 58, pp. 236–246. doi:10.1016/j.jtrangeo.2016.12.016.
9. Keeling, D.J. (2008) 'Transportation geography - new regional mobilities', *Progress in Human Geography*, 32(2), pp. 275–283. doi:10.1177/0309132507084400.
10. Mora, R., Truffello, R. and Oyarzún, G. (2021) 'Equity and accessibility of cycling infrastructure: An analysis of Santiago de Chile', *Journal of Transport Geography*, 91, p. N.PAG. doi:10.1016/j.jtrangeo.2021.102964.
11. Wolfe, R.I. (1962) 'Transportation and Politics: The Example of Canada', *Annals of the Association of American Geographers*, 52(2), pp. 176–190. doi:10.1111/j.1467-8306.1962.tb00404.x.